

Cloud-Based Machine Learning

Comprehensive Lecture Notes

By

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Module Focus: AWS · Google Cloud Platform · Microsoft Azure · IBM Cloud

Module Overview

This module covers concepts of Software-as-a-Service, Platform-as-a-Service, and Infrastructure-as-a-Service. Various cloud services for machine learning — including AWS, GCP, Microsoft Azure, and IBM Cloud — are explored and compared. Popular machine learning algorithms such as Linear Regression, Logistic Regression, Support Vector Machine, and Artificial Neural Networks are used to train models in a cloud IDE such as Google Colab. In this process, available public machine learning datasets are explored, and accessing them via cloud services is explained and applied through hands-on exercises.

Graduate Attributes & Learning Outcomes

A. Knowledge and Understanding

Having successfully completed this module, students should be able to demonstrate knowledge and understanding of:

- i. Types of services such as SaaS, IaaS, and PaaS
- ii. Advantages of using cloud services for machine learning
- iii. Major cloud services for ML: AWS, GCP, Microsoft Azure, and IBM Cloud
- iv. Requirements for using cloud services for machine learning
- v. Available public datasets for machine learning
- vi. Difference between pre-trained and custom ML models
- vii. Free cloud machine learning IDE such as Google Colab

B. Cognitive / Intellectual Skills / Application of Knowledge

- viii. Configure and install packages in a cloud environment
- ix. Import data to a cloud machine learning environment
- x. Train and evaluate a machine learning model in a cloud environment

C. Transferable / Key Skills

- xi. Apply machine learning in different areas such as medicine, banking, governance and more
- xii. Use acquired knowledge and skills in their career independently or with little guidance

D. Global Citizenship & Ethical Awareness

- xiii. Use cloud platforms to train ML models to solve problems with global impact
- xiv. Develop awareness of potential problems in other countries solvable using ML
- xv. Develop ML models in the cloud in accordance with common guidelines and ethical standards

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PART I — CLOUD COMPUTING FOUNDATIONS

1. Introduction to Cloud Computing & Machine Learning

1.1 What Is Cloud Computing?

Cloud computing is the on-demand delivery of computing resources — servers, storage, databases, networking, software, analytics, and intelligence — over the Internet ("the cloud"). Instead of owning and maintaining physical data centres, organisations rent access to anything from applications to storage from a cloud service provider (CSP) and pay only for what they use.

1.2 What Is Machine Learning?

Machine learning (ML) is a sub-field of artificial intelligence (AI) that enables computers to learn patterns and make decisions from data without being explicitly programmed for every scenario. An ML system improves its performance on a task as it is exposed to more data over time.

The three canonical learning paradigms are:

- **Supervised Learning** — The model is trained on labelled data (input–output pairs). Examples: classification, regression.
- **Unsupervised Learning** — The model finds hidden patterns in unlabelled data. Examples: clustering, dimensionality reduction.
- **Reinforcement Learning** — An agent learns by interacting with an environment and receiving rewards or penalties.

1.3 Why Combine Cloud Computing with Machine Learning?

Training ML models — especially deep neural networks — requires significant computational power (GPUs/TPUs), large storage, and specialised software stacks. Cloud platforms provide all of these on-demand, removing the need for expensive on-premises hardware and enabling rapid experimentation at scale.

Module Aim: *This course equips you with the knowledge and skills to use available cloud platforms (AWS, GCP, Microsoft Azure) to teach a computer to learn concepts from data without being explicitly programmed.*

2. Cloud Service Models: SaaS, PaaS & IaaS

Cloud services are categorised by how much the provider manages on your behalf versus how much you control yourself. The three principal models sit on a spectrum from fully managed (SaaS) to self-managed (IaaS).

2.2 Infrastructure-as-a-Service (IaaS)

IaaS provides virtualised computing resources: VMs, storage, networking. The user manages OS and above; the provider manages physical hardware.

- **Examples:** Amazon EC2, Azure VMs, Google Compute Engine.
- **ML Use Case:** Spinning up a GPU VM to train a custom TensorFlow model.

2.3 Platform-as-a-Service (PaaS)

PaaS provides a managed platform (runtime, middleware, OS). Developers focus only on application code and data.

- **Examples:** AWS Elastic Beanstalk, Google App Engine, Heroku.

2.4 Software-as-a-Service (SaaS)

SaaS delivers fully managed, ready-to-use applications. The provider manages everything.

- **Examples:** Google Workspace, Salesforce, Dropbox, Microsoft 365.

2.5 Comparison Summary

Aspect	IaaS	PaaS	SaaS
You Manage	Apps, Data, Runtime, MW, OS	Apps, Data	Nothing
Provider Manages	Virtualisation + below	Runtime, MW, OS + below	Everything
Flexibility	High	Medium	Low
Technical Skill	High	Medium	Low
Examples	AWS EC2, Azure VMs	Heroku, App Engine	Salesforce, Google Workspace

3. Advantages of Cloud Services for Machine Learning

#	Advantage	Explanation
1	On-Demand Scalability	Scale GPUs/TPUs up or down in minutes. Pay only while training.
2	No Upfront CapEx	No need to buy GPU servers. Converts CapEx to OpEx.
3	Specialised Hardware	Google TPUs, AWS

		Trainium/Inferentia, Azure confidential compute.
4	Managed ML & AutoML	SageMaker Autopilot, Vertex AutoML, Azure AutoML.
5	Collaboration	Cloud notebooks enable team sharing from any location.
6	MLOps Pipelines	Versioning, monitoring, CI/CD, automated retraining.
7	Pre-Built APIs	Vision, speech, language APIs (Rekognition, Vision AI, etc.).
8	Security & Compliance	ISO 27001, SOC 2, GDPR, HIPAA certifications.
9	Global Availability	Deploy models close to users via worldwide regions.
10	Free Tiers	AWS Educate, Google for Students, Azure for Students.

⚠ Drawbacks: Vendor lock-in, data transfer costs, internet dependency, variable pricing, data privacy/sovereignty concerns.

4. Major Cloud ML Platforms: AWS, GCP, Azure & IBM Cloud

4.1 AWS — SageMaker

AWS holds ~31% of the global cloud market (2026). SageMaker (2017) provides end-to-end ML: Studio, Autopilot, Canvas, Feature Store, Pipelines, Clarify. Pre-trained APIs: Rekognition, Comprehend, Transcribe, Translate.

4.2 GCP — Vertex AI

GCP holds ~12% but leads in AI/ML. Vertex AI (2021): Workbench, AutoML, Pipelines, Model Registry, BigQuery ML. Native TensorFlow and TPU access. \$300 free credits.

4.3 Azure — Azure Machine Learning

Azure holds ~25%. Azure ML: Studio, Designer (drag-and-drop), AutoML, Pipelines. Cognitive Services, Azure OpenAI Service. Best no-code experience. \$200 free trial.

4.4 IBM Cloud — Watson Studio

IBM focuses on enterprise governance. Watson Studio, Watson ML, AutoAI, OpenPages. Hybrid/multi-cloud via Cloud Pak. Free Lite plan.

4.5 Platform Comparison

Feature	AWS SageMaker	Google Vertex AI	Azure ML	IBM Watson
Market Share	~31%	~12%	~25%	~4%
AutoML	Autopilot	Vertex AutoML	Azure AutoML	AutoAI
No-Code	Canvas	AutoML Console	ML Designer	AutoAI GUI
Hardware	Trainium, Inferentia	TPU v5p/v6	Intel SGX	GPUs
Free Tier	250 hrs / 2 mo	\$300 / 90 days	\$200 / 30 days	Lite plan
Responsible AI	Clarify	Explainable AI	Fairness tools	OpenPages

5. Requirements for Using Cloud ML Services

5.1 Account & Access

- **Cloud Account:** Create a free-tier or student account on your chosen platform.
- **IAM:** Set up roles and permissions — follow the principle of least privilege.
- **Billing:** Set budget alerts to avoid unexpected charges.

5.2 Data Requirements

- Sufficient, relevant, high-quality data (hundreds to thousands of labelled examples).
- Data cleaning & preprocessing (missing values, outliers, encoding, normalisation).
- Cloud storage (S3, Cloud Storage, Blob Storage).

5.3 Software & Libraries

- **Language:** Python (dominant), R, SQL (BigQuery ML).
- **Frameworks:** Scikit-learn, TensorFlow, PyTorch, XGBoost.
- **Cloud SDKs:** Boto3 (AWS), Google Cloud SDK, Azure SDK.

5.4 Connectivity & Compute

- Stable Internet connection.
- Appropriate instance type (CPU, GPU, or TPU) based on model and dataset size.

6. Public Datasets for Machine Learning

Repository	Description	Typical Use Cases
Kaggle Datasets	Thousands of datasets, notebooks, competitions	Classification, NLP, image recognition
UCI ML Repository	Oldest academic source; clean, documented	Benchmarking, core algorithms
Google Dataset Search	Search engine for datasets worldwide	Domain-specific discovery
AWS Open Data	Large-scale (genomics, satellite, weather)	Big data, geospatial, life sciences
Hugging Face	NLP, vision, audio with Transformers	NLP, image classification
TensorFlow Datasets	MNIST, CIFAR, IMDB optimised for TF	Deep learning
Government Open Data	data.gov, data.gov.uk, data.gov.za	Policy, social science
OpenML	Collaborative ML benchmarking	Algorithm comparison

Classic Benchmarks: MNIST (70K digit images), CIFAR-10/100, Iris (150 samples), California Housing, IMDB Reviews (50K), ImageNet (14M+ images).

7. Pre-Trained vs Custom ML Models

7.1 Pre-Trained Models

A pre-trained model has already been trained on a massive dataset. You use it directly or fine-tune it on your smaller dataset.

Domain	Model	Creator	Use Case
NLP	BERT	Google	Text classification, Q&A
NLP	GPT-4	OpenAI	Text generation, chat
Vision	ResNet	Microsoft	Image classification
Vision	YOLO	Ultralytics	Object detection
Speech	Whisper	OpenAI	Speech-to-text (99 languages)
Science	AlphaFold	DeepMind	Protein structure prediction

7.2 Custom Models

A custom model is designed and trained from scratch on domain-specific data — essential when pre-trained models lack accuracy for specialised tasks.

7.3 Transfer Learning

Transfer learning starts with a pre-trained model and fine-tunes it on your dataset. The base layers act as feature extractors; the top layers are retrained for your specific task.

7.4 Comparison

Aspect	Pre-Trained	Custom	Transfer Learning
Data Needed	None / small	Large, domain-specific	Small–Medium
Compute Cost	Low	High	Medium
Time to Deploy	Hours	Weeks–Months	Days
Domain Accuracy	Good (general)	Best (if data sufficient)	Very Good
Expertise	Low–Medium	High	Medium

8. Free Cloud ML IDEs — Google Colab & Alternatives

8.1 Google Colab

Google Colab is a free, cloud-hosted Jupyter notebook that requires zero configuration and provides free GPU/TPU access.

- **Free GPU/TPU:** NVIDIA T4 GPUs free; A100s on paid plans (\$11.99/mo Pro, \$49.99/mo Pro+).
- **Pre-installed:** NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow, PyTorch, Keras.
- **Drive Integration:** Auto-save to Google Drive; mount Drive for persistent files.
- **AI Assistance:** Gemini-powered code completions.
- **Limitations:** Session timeouts (~90 min idle, ~12 hrs max). Not for production.

8.2 Alternatives

Platform	Free GPU?	Key Feature
Kaggle Notebooks	Yes (30 hrs/week)	Kaggle datasets integration
SageMaker Studio Lab	Yes (limited)	Free Jupyter on AWS

Gradient (Paperspace)	Yes (limited)	Free GPU notebooks
Lightning AI Studios	Yes (credits)	PyTorch Lightning integration

PART II — CORE ML ALGORITHMS

9. Linear Regression

9.1 What Is Linear Regression?

Linear regression is a supervised learning algorithm for predicting a continuous numerical output. It assumes a linear relationship between inputs and output:

- Simple: $\hat{y} = \beta_0 + \beta_1 x$
- Multiple: $\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$

Where $\hat{y}$ is predicted value, β_0 is intercept, and $\beta_1 \dots \beta_n$ are learned coefficients.

9.2 How It Learns — Ordinary Least Squares

The model finds the best-fit line by minimising the Sum of Squared Errors (SSE): $SSE = \sum (y_i - \hat{y}_i)^2$. The algorithm adjusts coefficients until SSE is minimised.

9.3 Evaluation Metrics for Regression

Metric	Formula	Interpretation
MAE	$\sum y_i - \hat{y}_i / n$	Average absolute error. Robust to outliers.
MSE	$\sum (y_i - \hat{y}_i)^2 / n$	Average squared error. Penalises large errors.
RMSE	$\sqrt{\text{MSE}}$	Same unit as target. Most commonly reported.
R ² Score	$1 - (SS_{\text{res}} / SS_{\text{tot}})$	Proportion of variance explained. 1.0 = perfect.

9.4 Hands-On: Linear Regression in Google Colab

```
from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
import numpy as np

data = fetch_california_housing()
X, y = data.data, data.target
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```

model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print(f"RMSE: {np.sqrt(mean_squared_error(y_test, y_pred)):.4f}")
print(f"R²: {r2_score(y_test, y_pred):.4f}")

```

9.5 Assumptions & Limitations

- Assumes a linear relationship between features and target.
- Sensitive to outliers.
- Assumes no multicollinearity among features.
- Not suitable for classification tasks.

Applications: House price prediction, sales forecasting, medical cost estimation, stock trends, crop yield prediction.

10. Logistic Regression

10.1 What Is Logistic Regression?

Despite its name, logistic regression is a classification algorithm. It predicts the probability that an input belongs to a particular class using the sigmoid function:

$$\sigma(z) = 1 / (1 + e^{-z}) \text{ where } z = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n$$

If probability $\geq 0.5 \rightarrow$ Class 1; otherwise $\rightarrow$ Class 0.

10.2 Binary vs Multi-Class

- **Binary:** Two classes (e.g., benign vs malignant). Single sigmoid.
- **Multi-class:** Multiple classes. Trains one classifier per class (One-vs-Rest).
- **Softmax:** Generalisation for multi-class; outputs probability distribution.

10.3 Cost Function — Log Loss

$J = -(1/n) \sum [y_i \log(\hat{y}_i) + (1-y_i) \log(1-\hat{y}_i)]$. Optimised via gradient descent.

10.4 Hands-On: Logistic Regression in Google Colab

```

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import classification_report

data = load_breast_cancer()
X, y = data.data, data.target
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42,
stratify=y)
scaler = StandardScaler()
X_train_s = scaler.fit_transform(X_train)
X_test_s = scaler.transform(X_test)

```

```

model = LogisticRegression(max_iter=5000)
model.fit(X_train_s, y_train)
print(classification_report(y_test, model.predict(X_test_s), target_names=data.target_names))

```

Applications: Medical diagnosis, spam detection, credit default prediction, customer churn, sentiment analysis.

11. Support Vector Machines (SVM)

11.1 What Is an SVM?

A Support Vector Machine finds the optimal hyperplane that separates classes with the maximum margin. Support vectors are the data points closest to the hyperplane — they define its position.

11.2 The Kernel Trick

When data is not linearly separable, SVM uses a kernel function to project data into a higher-dimensional space where a linear separator exists.

Kernel	Use Case	Scikit-learn Parameter
Linear	Linearly separable data	kernel='linear'
RBF	Complex boundaries (default)	kernel='rbf'
Polynomial	Polynomial boundaries	kernel='poly'
Sigmoid	Neural network-like	kernel='sigmoid'

11.3 SVM vs Logistic Regression

Aspect	Logistic Regression	SVM
Objective	Maximise probability	Maximise margin
Output	Probabilities (0–1)	Class label
Outlier Sensitivity	High	Low (only support vectors)
Non-Linear Data	Needs feature engineering	Kernel trick handles it

11.4 Hands-On: SVM in Google Colab

```

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import classification_report

```

```

iris = load_iris()
X, y = iris.data, iris.target
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42,
stratify=y)
scaler = StandardScaler()
X_train_s = scaler.fit_transform(X_train)
X_test_s = scaler.transform(X_test)

svm = SVC(kernel='rbf', C=1.0, gamma='scale', random_state=42)
svm.fit(X_train_s, y_train)
print(classification_report(y_test, svm.predict(X_test_s), target_names=iris.target_names))

```

Applications: Image classification, handwriting recognition, cancer detection, face recognition, text categorisation.

12. Artificial Neural Networks (ANN)

12.1 What Is an ANN?

An ANN is a computing system inspired by the brain. It consists of interconnected neurons in layers that learn to map inputs to outputs by adjusting weights through training.

- **Input Layer:** Receives feature values (one neuron per feature).
- **Hidden Layer(s):** Perform computations using weights, biases, and activation functions.
- **Output Layer:** Produces the final prediction.

12.2 How an ANN Learns — Backpropagation

1. Forward Pass: Data flows through layers; each neuron computes weighted sum + bias + activation.
2. Loss Computation: Difference between predicted and true values.
3. Backward Pass: Loss gradient propagated backwards; weights updated proportionally.
4. Weight Update: Using optimiser (SGD, Adam). Repeat for many epochs.

12.3 Common Activation Functions

Function	Formula	Range	Typical Use
Sigmoid	$1 / (1 + e^{-z})$	(0, 1)	Binary output
Tanh	$(e^z - e^{-z}) / (e^z + e^{-z})$	(-1, 1)	Hidden layers
ReLU	$\max(0, z)$	$[0, \infty)$	Most common hidden
Softmax	$e^{z_i} / \sum e^{z_j}$	(0,1) sums to 1	Multi-class output

12.4 Hands-On: ANN in Google Colab

```
import tensorflow as tf
from tensorflow.keras import layers, models

(X_train, y_train), (X_test, y_test) = tf.keras.datasets.mnist.load_data()
X_train = X_train.reshape(-1, 784).astype('float32') / 255.0
X_test = X_test.reshape(-1, 784).astype('float32') / 255.0

model = models.Sequential([
    layers.Dense(128, activation='relu', input_shape=(784,)),
    layers.Dropout(0.2),
    layers.Dense(64, activation='relu'),
    layers.Dropout(0.2),
    layers.Dense(10, activation='softmax')
])
model.compile(optimizer='adam', loss='sparse_categorical_crossentropy', metrics=['accuracy'])
model.fit(X_train, y_train, epochs=10, batch_size=128, validation_split=0.1)
test_loss, test_acc = model.evaluate(X_test, y_test)
print(f"Test Accuracy: {test_acc:.4f}")
```

12.5 Algorithm Comparison Summary

Aspect	Linear Reg.	Logistic Reg.	SVM	ANN
Task	Regression	Classification	Classification (+Reg)	Both
Output	Continuous	Probability → class	Class label	Continuous/prob
Complexity	Low	Low	Medium	High
Interpretability	High	High	Medium	Low (black box)
Non-Linearity	No	No	Yes (kernel)	Yes (inherently)
Training Speed	Fast	Fast	Moderate	Slow

PART III — HANDS-ON CLOUD ML SKILLS

13. Configuring a Cloud ML Environment

13.1 Setting Up Google Colab

```
# Check libraries and GPU
!pip list
!python --version
import torch
print("GPU available:", torch.cuda.is_available())

# Install packages
!pip install xgboost==2.1.0 lightgbm catboost shap

# Mount Google Drive
from google.colab import drive
drive.mount('/content/drive')
```

13.2 Setting Up AWS SageMaker

```
!pip install sagemaker --upgrade
import sagemaker
session = sagemaker.Session()
bucket = session.default_bucket()
role = sagemaker.get_execution_role()
```

13.3 Setting Up Azure ML

```
!pip install azureml-core azureml-train-automl
from azureml.core import Workspace
ws = Workspace.from_config()
print("Workspace:", ws.name)
```

Best Practice: Pin library versions in `requirements.txt` for reproducibility.

14. Importing Data into a Cloud ML Environment

14.1 Google Colab — Multiple Methods

```
# Option A: Upload from local machine
from google.colab import files
uploaded = files.upload()

# Option B: From Google Drive
from google.colab import drive
drive.mount('/content/drive')
```

```
df = pd.read_csv('/content/drive/MyDrive/data.csv')

# Option C: From URL
df = pd.read_csv("https://raw.githubusercontent.com/.../iris.csv")

# Option D: Built-in datasets
from sklearn.datasets import load_iris
iris = load_iris()
```

14.2 AWS S3 & Azure Blob Storage

```
# AWS S3
import boto3; from io import StringIO
s3 = boto3.client('s3')
obj = s3.get_object(Bucket='my-bucket', Key='data/train.csv')
df = pd.read_csv(StringIO(obj['Body'].read().decode('utf-8')))

# Azure Blob
from azureml.core import Workspace, Dataset
ws = Workspace.from_config()
dataset = Dataset.Tabular.from_delimited_files(
    path=(ws.get_default_datastore(), 'data/train.csv'))
df = dataset.to_pandas_dataframe()
```

15. Training & Evaluating ML Models in the Cloud

15.1 Multi-Algorithm Comparison on Iris

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score
import tensorflow as tf

iris = load_iris(); X, y = iris.data, iris.target
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42,
    stratify=y)
scaler = StandardScaler()
X_train_s = scaler.fit_transform(X_train)
X_test_s = scaler.transform(X_test)

# Logistic Regression
lr = LogisticRegression(max_iter=200).fit(X_train_s, y_train)
print("LR:", accuracy_score(y_test, lr.predict(X_test_s)))

# SVM
svm = SVC(kernel='rbf').fit(X_train_s, y_train)
print("SVM:", accuracy_score(y_test, svm.predict(X_test_s)))

# ANN
ann = tf.keras.Sequential([
```

```

tf.keras.layers.Dense(64, activation='relu', input_shape=(4,)),
tf.keras.layers.Dense(32, activation='relu'),
tf.keras.layers.Dense(3, activation='softmax'))
ann.compile(optimizer='adam', loss='sparse_categorical_crossentropy', metrics=['accuracy'])
ann.fit(X_train_s, y_train, epochs=50, batch_size=16, verbose=0)
_, acc = ann.evaluate(X_test_s, y_test, verbose=0)
print("ANN:", round(acc, 4))

```

15.2 Key Evaluation Metrics

Metric	Type	Best For
Accuracy	Classification	Balanced classes
Precision	Classification	When FP are costly
Recall	Classification	When FN are costly (medical)
F1-Score	Classification	Imbalanced classes
AUC-ROC	Classification	Binary discriminative ability
RMSE	Regression	Penalises large errors
R² Score	Regression	Variance explained

PART IV — APPLICATIONS, GLOBAL IMPACT & ETHICS

16. ML Applications: Medicine, Banking, Governance & Beyond

Machine learning is a tool for solving real-world problems. As Level 3 students preparing for professional careers, you should identify opportunities to apply ML across diverse domains.

16.1 Medicine & Healthcare

Application	ML Technique	Example
Medical Image Diagnosis	CNNs	Tumour detection in X-rays, MRIs, CT scans
Disease Risk Prediction	Logistic Reg., RF, ANN	Predicting diabetes from patient records
Drug Discovery	Deep Learning, RL	AlphaFold for protein structure prediction
Electronic Health Records	NLP, Classification	Extracting diagnoses from clinical notes
Hospital Operations	Regression, Time-Series	Predicting admission volumes

16.2 Banking & Finance

Application	ML Technique	Example
Fraud Detection	Anomaly Detection, SVM, ANN	Real-time suspicious transaction identification
Credit Scoring	Logistic Reg., Gradient Boost	Loan default prediction
Algorithmic Trading	RL, LSTM	Automated trading strategies
Customer Segmentation	K-Means, DBSCAN	Targeted marketing
Anti-Money Laundering	Graph NN, Anomaly Det.	Complex laundering pattern detection

16.3 Governance & Public Sector

Application	ML Technique	Example
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Public Health Surveillance	NLP, Anomaly Detection	Disease outbreak tracking
Tax Revenue Prediction	Linear Reg., Time-Series	Government budget planning
Smart Cities	IoT + ML, Computer Vision	Traffic optimisation, energy management
Citizen Service Chatbots	NLP, Conversational AI	Automating government enquiries

16.4 Other Domains

Domain	Application Examples
Agriculture	Crop disease detection, yield prediction, precision irrigation
Education	Adaptive learning, automated grading, dropout prediction
Transportation	Self-driving vehicles, route optimisation, demand forecasting
Retail & E-Commerce	Recommendation engines, dynamic pricing, sentiment analysis
Energy	Smart grid, renewable forecasting, predictive maintenance
Environment	Climate modelling, deforestation monitoring, biodiversity tracking

17. ML for Global Impact & Solving Problems Across Borders

As global citizens, it is important to recognise that ML can address pressing challenges affecting millions worldwide. Cloud platforms democratise access — a student with a laptop and internet can build globally relevant solutions.

17.1 ML and the UN Sustainable Development Goals

Research shows AI/ML can benefit 82% of targets across the 17 UN SDGs:

SDG	ML Application	Example
SDG 1: No Poverty	Satellite imagery + classifiers	Mapping poverty from nighttime lights
SDG 2: Zero Hunger	Crop yield prediction	Weather/soil ML models in India

SDG 3: Good Health	Disease diagnostics	Malaria detection from blood smear images
SDG 4: Quality Education	Adaptive learning	AI tutoring for under-resourced schools
SDG 6: Clean Water	Sensor monitoring + ML	Predicting contamination in water supply
SDG 7: Clean Energy	Smart grid optimisation	Matching demand to solar/wind supply
SDG 11: Sustainable Cities	Traffic management	AI traffic control in Lagos, Nairobi
SDG 13: Climate Action	Climate modelling	Neural networks forecasting carbon emissions

17.2 Problems in Other Countries Solvable with ML

- **Sub-Saharan Africa:** Limited diagnostic facilities → ML-powered mobile diagnostics (retinal scans on smartphones).
- **South/Southeast Asia:** Flood prediction using time-series ML models on historical weather data.
- **Latin America:** Deforestation monitoring in the Amazon via satellite imagery classification.
- **Small Island States:** Coral reef health monitoring via underwater image classification.
- **Global:** NLP models for low-resource languages to preserve cultural heritage.

17.3 Industry Commitments

- **Google:** \$10M+ for AI projects addressing natural disasters and climate change.
- **Microsoft:** AI for Good lab in Africa, South America, North America; 2M people trained in AI.
- **OpenAI:** \$1M+ in financial and API commitments for SDG-focused organisations.
- **AWS:** Open Data Registry: free access to massive datasets for research.

18. Ethics, Guidelines & Responsible ML Development

Developing ML models comes with significant responsibility. Models can encode biases, violate privacy, and make life-altering decisions. This chapter ensures you develop models in accordance with ethical standards.

18.1 The Six Principles of Responsible AI

#	Principle	Description	Practical Action
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1	Fairness	Treat all groups equitably	Audit data; test across demographics
2	Transparency	Stakeholders understand decisions	Use LIME/SHAP; publish model cards
3	Accountability	Clear responsibility for outcomes	Governance committees; audit trails
4	Privacy	Protect sensitive data	Anonymise; GDPR/POPIA compliance
5	Robustness	Reliable and secure	Test adversarial attacks; monitor drift
6	Human Oversight	Humans in control for high-stakes	Human-in-the-loop frameworks

18.2 Sources of Bias in ML

- **Historical Bias:** Training data reflects past societal inequalities.
- **Representation Bias:** Under-representation of certain groups in training data.
- **Measurement Bias:** Features used as proxies correlate with protected attributes.
- **Aggregation Bias:** Single model applied to diverse populations.
- **Evaluation Bias:** Testing on benchmarks that don't represent deployment population.

🔗 Case Study: A US healthcare algorithm assigned lower risk scores to Black patients than equally sick White patients, because it used spending as a proxy for illness — Black patients had historically lower spending due to systemic barriers. The fix: redefine the target variable to use actual health indicators.

18.3 Regulatory Frameworks

Framework	Region	Key Requirements
EU AI Act	EU	Risk-based; bans unacceptable AI; strict rules for high-risk
GDPR	EU	Consent, right to explanation, data minimisation
POPIA	South Africa	Lawful processing, data subject rights, security safeguards
NIST AI RMF	USA	Voluntary: Govern, Map, Measure, Manage

Japan AI Basic Act	Japan	Risk-based; fairness audits; mandatory record-keeping
South Korea AI Act	S. Korea	Non-discrimination; AI content labelling; fines

18.4 Practical Ethical Guidelines for Your Projects

- Document your data: Record sources, methods, preprocessing, exclusions.
- Check for bias before training: Analyse class balance, demographic representation.
- Use explainability tools: SHAP or LIME to understand feature importance.
- Evaluate across subgroups: Report metrics by gender, age, ethnicity.
- Publish a model card: Document purpose, data, metrics, limitations, ethics.
- Obtain informed consent: Ensure subjects understand data usage.
- Plan for monitoring: Models degrade (data drift). Plan retraining.
- Consider environmental impact: Large training has a carbon footprint.

18.5 Cloud Platform Responsible AI Tools

Platform	Tool	Function
AWS	SageMaker Clarify	Bias detection; feature importance
GCP	Vertex Explainable AI	Feature attributions; What-If Tool
Azure	Responsible AI Dashboard	Fairness, error analysis, interpretability
IBM	AI Fairness 360 / OpenPages	Open-source bias toolkit; governance

19. Summary, Key Terms & Further Reading

19.1 Module Summary

This module covered the complete landscape of cloud-based machine learning across four parts:

Part I: Cloud foundations — service models, advantages, platforms, datasets, pre-trained vs custom models, free IDEs.

Part II: Core algorithms — Linear Regression, Logistic Regression, SVM, and ANN with theory and code.

Part III: Hands-on skills — configuring environments, importing data, training and evaluating models.

Part IV: Applications across medicine, banking, governance; global impact and SDGs; ethics and responsible AI.

19.2 Key Terms Glossary

Cloud Computing: On-demand delivery of computing resources over the Internet.

IaaS / PaaS / SaaS: Three cloud service models with increasing provider management.

Linear Regression: Supervised algorithm predicting continuous output via linear equation.

Logistic Regression: Classification algorithm using sigmoid function for probabilities.

SVM: Classification algorithm finding maximum-margin hyperplane.

ANN: Brain-inspired model of interconnected neurons in layers.

Backpropagation: Algorithm computing loss gradients to update neural network weights.

Activation Function: Non-linear function applied to neuron output (ReLU, sigmoid, softmax).

Kernel Trick: Transformation projecting data to higher dimensions for separability.

AutoML: Tools automating model selection and hyperparameter tuning.

Pre-Trained Model: Model already trained on large dataset, ready for use or fine-tuning.

Transfer Learning: Fine-tuning a pre-trained model on a new, smaller dataset.

Google Colab: Free cloud-hosted Jupyter notebook with GPU/TPU access.

Bias (in ML): Systematic error leading to unfair outcomes for certain groups.

Explainability: Ability to understand how a model makes predictions.

SHAP / LIME: Tools providing feature-level explanations for predictions.

Model Card: Documentation of model purpose, data, metrics, limitations, ethics.

GDPR / POPIA: Data protection regulations (EU and South Africa).

EU AI Act: First comprehensive AI regulation with risk-based classification.

Confusion Matrix: Table showing TP, TN, FP, FN for classification evaluation.

F1-Score: Harmonic mean of precision and recall.

R² Score: Proportion of variance explained by regression model.

19.3 Further Reading & Resources

- AWS: aws.amazon.com/sagemaker
- GCP: cloud.google.com/vertex-ai
- Azure: learn.microsoft.com/azure/machine-learning
- IBM: cloud.ibm.com/catalog/services/watson-studio
- Google Colab: colab.research.google.com
- Kaggle Learn: kaggle.com/learn — Free micro-courses
- Scikit-learn: scikit-learn.org/stable
- TensorFlow: tensorflow.org/tutorials
- SHAP: shap.readthedocs.io — Model explainability
- AI Fairness 360: aif360.mybluemix.net — IBM bias toolkit
- Géron, A. (2022). Hands-On ML with Scikit-Learn, Keras, and TensorFlow. O'Reilly.
- Bishop, C. & Bishop, H. (2023). Deep Learning: Foundations and Concepts. Springer.
- Barocas, S. et al. (2023). Fairness and Machine Learning. MIT Press.

Appendix: Complete Learning Outcomes Mapping

Category	LO	Learning Outcome	Chapter
A. Knowledge	i	Types of services: SaaS, IaaS, PaaS	Ch 2
A. Knowledge	ii	Advantages of cloud services for ML	Ch 3
A. Knowledge	iii	Major cloud services	Ch 4
A. Knowledge	iv	Requirements for cloud ML	Ch 5
A. Knowledge	v	Public datasets for ML	Ch 6
A. Knowledge	vi	Pre-trained vs custom models	Ch 7
A. Knowledge	vii	Free cloud ML IDEs	Ch 8
B. Cognitive	viii	Configure & install packages	Ch 13
B. Cognitive	ix	Import data to cloud ML	Ch 14
B. Cognitive	x	Train & evaluate ML models	Ch 9–12, 15
C. Transferable	xi	Apply ML in medicine, banking, etc.	Ch 16
C. Transferable	xii	Use skills independently in career	Ch 13–16
D. Global/Ethics	xiii	Solve problems with global impact	Ch 17
D. Global/Ethics	xiv	Awareness of global problems	Ch 17
D. Global/Ethics	xv	Ethical ML development	Ch 18